Stone Tao

stonezt2019@gmail.com | github.com/stonet2000 | stoneztao.com | stoneztao.substack.com

Education

PhD Student in Computer Science and Engineering 2023 – 2026
University of California, San Diego. Advisors: Hao Su, Henrik Christensen
Research Topics: Simulation, Robotics, Reinforcement Learning, Evaluation

B.S. Double Major in Computer Science and Cognitive Science, Math Minor 2019 – 2023
University of California, San Diego.

Experience

Physical Intelligence, Member of Technical Staff 2026 – Present
- Research on simulation for robotics foundation models.

Allen Institute for AI (AI2), Research Intern, Advisor: Dieter Fox 2026
- Research on scalable simulation infrastructure, evaluations, and reinforcement learning for manipulation.

NVIDIA Research, Research Intern, Advisors: Iretiayo Akinola, Yijie Guo, Yashraj Narang 2025
- Research on sim-to-real, reinforcement learning, simulation, and manipulation.

Hillbot/Sudo AI, Research Scientist Intern, Advisor: Hao Su 2024
- Research on the intersection of simulation, reinforcement learning, imitation learning, and robotics.

Lux AI Challenge, Co-Founder 2021 – Present
- Lead the research, development, and design of creative, accessible, and inclusive AI competitions.

QuantCo, ML Engineer Intern, Advisor: T. Ben Thompson 2021
- Research on high-precision and accurate function approximation via deep learning and decision trees.

Selected Awards

National Science Foundation Graduate Research Fellowship 2023
UCSD CSE Award for Excellence in Leadership 2023

Publications / Preprints

Staggered Environment Resets Improve Massively Parallel On-Policy Reinforcement Learning
Neural Information and Processing Systems **(NeurIPS)** 2025.
Sid Bharthulwar, **Stone Tao**, Hao Su.
arXiv

Multi-Stage Manipulation with Demonstration-Augmented Reward, Policy, and World Model Learning
International Conference on Machine Learning **(ICML)** 2025.
Adrià López Escoriza, Nicklas Hansen, **Stone Tao**, Tongzhou Mu, Hao Su.
arXiv | project page | code

Do Vision-Language Models Have Internal World Models? Towards an Atomic Evaluation
Association for Computational Linguistics **(ACL)** 2025.
Qiyue Gao*, Xinyu Pi*, Kevin Liu, Junrong Chen, Ruolan Yang, Xinqi Huang, Xinyu Fang, Lu Sun, Gautham Kishore, Bo Ai, **Stone Tao**, Mengyang Liu, Jiaxi Yang, Chao-Jung Lai, Chuanyang Jin, Jiannan Xiang, Benhao Huang, Zeming Chen, David Danks, Hao Su, Tianmin Shu, Ziqiao Ma, Lianhui Qin, Zhiting Hu.
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ManiSkill-HAB: A Benchmark for Low-Level Manipulation in Home Rearrangement Tasks
International Conference on Learning Representations **(ICLR)** 2025.
Arth Shukla, **Stone Tao**, Hao Su.
arXiv | project page | code

Policy Decorator: Model-Agnostic Online Refinement for Large Policy Model
International Conference on Learning Representations **(ICLR)** 2025.
Xiu Yuan*, Tongzhou Mu*, **Stone Tao**, Yunhao Fang, Michael Zhang, Hao Su.
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ManiSkill3: GPU Parallelized Robotics Simulation and Rendering for Generalizable Embodied AI
Robotics Science and Systems **(RSS)** 2025
Robot Learning Workshop at **(ICLR)** 2025, Oral Presentation
Stone Tao, Fanbo Xiang, Arth Shukla, Yuzhe Qin, Xander Hinrichsen, Xiaodi Yuan, Chen Bao, Xinsong Lin, Yulin Liu, Tse-kai Chan, Yuan Gao, Xuanlin Li, Tongzhou Mu, Nan Xiao, Arnav Gurha, Viswesh Nagaswamy Rajesh, Yong Woo Choi, Yen-Ru Chen, Zhiao Huang, Roberto Calandra, Rui Chen, Shan Luo, Hao Su.
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Lux AI Season 3: Multi-Agent Meta Learning at Scale
Neural Information and Processing Systems **(NeurIPS)** 2024 Competitions Track.
Stone Tao, Akarsh Kumar, Bovard Doerschuk-Tiberi, Isabelle Pan, Addison Howard, Hao Su.
competition page | code

Reverse Forward Curriculum Learning for Extreme Sample and Demonstration Efficiency in RL
International Conference on Learning Representations **(ICLR)** 2024.
Stone Tao, Arth Shukla, Tse-kai Chan, Hao Su.
arXiv | project page | code

MetaWriter: Exploring the Potential and Perils of AI Writing Support in Scientific Peer Review
26th ACM Conference on Computer-Supported Cooperative Work and Social Computing **(CSCW)** 2024.
Lu Sun, **Stone Tao**, Junjie Hu, Steven Dow.
ACM Digital Library Link

Lux AI Season 2: NeurIPS Edition
Neural Information and Processing Systems **(NeurIPS)** 2023 Competitions Track.
Stone Tao, Qimai Li, Yuhao Jiang, Jiaxin Chen, Xiaolong Zhu, Bovard Doerschuk-Tiberi, Isabelle Pan, Addison Howard.
competition page

Abstract-to-Executable Trajectory Translation for One-Shot Task Generalization
International Conference on Machine Learning **(ICML)** 2023.
Stone Tao, Xiaochen Li, Tongzhou Mu, Zhiao Huang, Yuzhe Qin, Hao Su.
arXiv | project page | code

ManiSkill2: a Unified Benchmark for Generalizable Manipulation Skills
International Conference on Learning Representations **(ICLR)** 2023
Jiayuan Gu†, Fanbo Xiang†, Xuanlin Li*, Zhan Ling*, Xiqiang Liu*, Tongzhou Mu*, Yihe Tang*, **Stone Tao***, Xinyue Wei*, Yunchao Yao*, Xiaodi Yuan, Pengwei Xie, Zhiao Huang, Rui Chen, Hao Su.
arXiv | project page | code

Maniskill: Generalizable manipulation skill benchmark with large-scale demonstrations
Neural Information and Processing Systems **(NeurIPS)** 2021 Datasets and Benchmarks Track
Tongzhou Mu*, Zhan Ling*, Fanbo Xiang*, Derek Cathera Yang*, Xuanlin Li*, **Stone Tao**, Zhiao Huang, Zhiwei Jia, Hao Su.
arXiv | project page | code

Emergent Collective Intelligence From Massive-Agent Cooperation and Competition
Neural Information and Processing Systems **(NeurIPS)** 2022 Deep RL Workshop.
Hanmo Chen*, **Stone Tao***, Jiaxin Chen, Weihan Shen, Xihui Li, Sikai Cheng, Xiaolong Zhu, Xiu Li.
arXiv

Selected Open Source Projects

ManiSkill ⋆3000+
GPU parallelized robotics simulation framework and infrastructure. Designed to be pythonic and abstract away

most GPU memory management code to make building and training on robotics tasks a breeze.
mani-skill/ManiSkill

Lux AI Challenge ⋆1600+
Code for the Lux AI Challenge, including the AI environments and visualizers.
Lux-AI-Challenge

Presentations / Talks

Accelerating and Driving Robotics Progress via Simulation Integrated Robot Learning
University of Pennsylvania 2026
Youtube Recording

Scaling Embodied AI via Simulation and Sim-integrated Robot Learning
Allen AI (AI2) 2025
Institute for AI Industry Research (AIR), Tsinghua University 2024

ManiSkill3: Scalable Simulation+Rendering for Generalizable Embodied AI
Robopapers (Podcast) 2025
CVPR 2024 Embodied AI Workshop 2024
Echo AI Talk 2024
Youtube Recording

Service

Challenge Organization
Robotics: ManiSkill Challenge 2021 (ICLR 2022 Generalizable Policy Learning in the Physical World Workshop), ManiSkill2 Challenge (CVPR 2023 Embodied AI Workshop)
Multiagent Games/RL: Lux AI Season 1, Lux AI Season 2 (NeurIPS 2023 Competitions), Lux AI Season 3 (NeurIPS 2024 Competitions)

Peer Review
AI/ML: ICLR, ICML, NeurIPS (Top Reviewer), TMLR, CoLLas, RLC
Robotics: IROS, ICRA, RSS, CoRL
Computer Vision/Graphics: SIGGRAPH